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B.Tech Thesis

HUMANOID ROBOT WITH EDGE AI

Submitted by

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Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology
in
Electronics and Communication Engineering

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SUPERVISOR CERTIFICATE

This is to certify that the thesis entitled "HUMANOID ROBOT WITH EDGE AI" is a bonafide record of the project work carried out by

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in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering at Dayananda Sagar University, School of Engineering, Bengaluru.

This work has been carried out during the academic year 2025–2026 under the guidance of Mr. ABHINAV KARAN, Department of Electronics and Communication Engineering.

To the best of our knowledge, this thesis has not been submitted elsewhere for the award of any other degree or diploma.

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Declaration

We hereby declare that the thesis entitled “HUMANOID ROBOT WITH EDGE AI” is a Bonafide record of the work carried out by us in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Electronics and Communication Engineering at Dayananda Sagar University, School of Engineering, Bengaluru.

This work has been carried out under the guidance of Dr. Supervisor Name, Department of Electronics and Communication Engineering.

We further declare that this work has not been submitted previously, either in full or in part, to any other university or institution for the award of any degree or diploma.

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Abstract

This thesis presents the design and implementation of an autonomous human-tracking wheeled humanoid robot utilizing Edge AI and a distributed software architecture. The system is developed on a Raspberry Pi platform running **Ubuntu 22.04 with ROS 2 Humble**, ensuring modularity and real-time performance.
Unlike traditional systems that rely on containerization, this implementation uses a **native ROS 2 deployment**, reducing system overhead and improving latency. The perception layer uses a **Logitech Brio 100 USB camera**, enabling plug-and-play compatibility and stable video acquisition.
A **YOLOv8n model optimized using ONNX Runtime** is used for real-time human detection. The cognition layer implements a **PID controller** to generate velocity and direction commands based on target position. These commands are transmitted via serial communication to an **ESP32 microcontroller**, which performs motor actuation using PWM.
The system demonstrates efficient real-time tracking (~30 FPS) with low latency (~33 ms), proving that advanced humanoid robotics can be achieved using affordable hardware and optimized edge computing.

KEYWORDS
Humanoid Robotics, ROS 2 Humble, Edge AI, YOLOv8, PID Control, Raspberry Pi, ESP32, Ubuntu 22.04

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Chapter 1

Introduction

1.1 Motivation

The advancements made in robotics and AI have led to a greater need for systems that can operate in human-centric settings. Existing systems use cloud-based computing, which results in delay and privacy issues. The motivation behind this project is the need to incorporate Edge AI into these systems.

1.2 Scope of Research

The research aims to design a real-time human tracking system using a modular humanoid platform. The scope of the project includes four main elements:

- Vision system: Implementation of the YOLOv8n model fine-tuned with ONNX for localized inference on the Raspberry Pi.
- System design: Development of an efficient communication network using ROS 2 Humble for low latency.
- Control design: Designing and tuning a PID controller to provide smooth control actions based on vision input.
- Hardware integration: Designing an effective interface between the high-level and low-level controllers (Raspberry Pi and ESP32, respectively).

1.3 Objectives

The objectives of this study can be stated as follows:

1. Implementation of an optimized YOLOv8 algorithm in a Raspberry Pi 4/5 for the purpose of real-time detection and localization of human entities.
2. Development of a modular system software based on ROS 2, which separates perception algorithms, decision-making routines, and hardware connections.
3. Development of a tracking algorithm, which ensures that the detected human is kept centered within the robot’s field of vision by proper control of motors.
4. Provision of inter-platform communications (serial/USB) in order to command the ESP32 slave device in real time.

1.4 Thesis Organization

The dissertation is broken down into several chapters to facilitate a logical technical flow:

- Chapter 2: Literature Review – This chapter provides an overview of existing advancements in Edge AI technology, human tracking techniques, and Robot Operating System (ROS) evolution.
- Chapter 3: System Design and Methodology – The chapter will cover hardware considerations, the structure of the ROS 2 node, and the PID controller design process.
- Chapter 4: Implementation – It describes the software development environment configuration (Debian Trixie), model conversion to ONNX format, and ESP32 firmware programming.
- Chapter 5: Results and Analysis – It discusses tracking precision, FPS, and control loop latency results.
- Chapter 6: Conclusion and Future Work – In this chapter, conclusions will be drawn based on research outcomes, along with suggestions for future improvements such as incorporating 3D depth perception and speech recognition capability.

1.5 Summary

The conclusion from chapter one is that artificial intelligence must be localized when used in robotics. The scope of this research paper is covered in this chapter, whereby the implementation of a novel approach using modern technology in object detection and proven concepts in control theory, all within the ROS 2 environment, is shown to be feasible.

Chapter 2

Theoretical Background

2.1 Introduction

The theory needed to understand the incorporation of computer vision, distributed computing systems, and control theory into the humanoid robot is developed in this chapter. The entire project depends on the effective interrelationship between top-level perception and bottom-level action. The success of autonomous human detection requires fast image processing and multi-threaded communication using the Robot Operating System (ROS).

2.2 Previous Work

Research in humanoid robotics has evolved from static, pre-programmed movements to dynamic, vision-guided interactions.

On YOLOv8 and Real-Time Perception

Al-Sarayreh et al. (2024). *"Real-Time Vehicle and Person Detection Using YOLOv8-Nano for Intelligent Transportation Systems."* [1].

This paper specifically benchmarks the Nano version of YOLOv8 (which you are using) and proves it achieves >80% accuracy in real-time scenarios on low-power hardware.

On ONNX and Edge Optimization

- Edge AI Inference: Microsoft (2024). *"ONNX Runtime: Cross-Platform Performance for Machine Learning Models on ARM64."*[3].

It explains how the runtime uses the Pi's CPU instructions to double the inference speed.

To justify the selection of the proposed technical stack, a comparative study was conducted against existing methodologies commonly found in recent literature. Table 2.1 summarizes the key differences between the selected framework and traditional approaches.

Table: 2.1

Feature	Traditional Approach	Proposed Methodology	Advantage
Middleware	ROS 1 (Melodic/Noetic)	ROS 2 Humble	Decentralized architecture; no single point of failure (Master node).
Object Detection	Haar Cascades / YOLOv5	YOLOv8n (ONNX Optimized)	Improved accuracy-to-latency ratio; anchor-free detection logic.
OS Environment	Ubuntu 20.04 (LTS)	Ubuntu 22.04	Compatibility with latest Linux kernels and Pi camera stacks.
Inference Engine	Native PyTorch/TF	ONNX Runtime	Hardware-specific acceleration for ARM64 CPUs.
Deployment	Native Installation	Native ROS	Environment isolation and complete replicability.

2.3 Research Gap

Despite considerable study in autonomous robots, there are still some missing elements in the field that this project seeks to bridge:

1. Operating System and Modern Hardware Stack: The bulk of literature on the subject makes use of outdated Debian distributions and the older camera-v1 hardware stack. This problem is solved by implementing the tracking system under Ubuntu 22.04 with the use of the updated Logitech Brio hardware stack.
2. Affordable Humanoid Robots: A number of high-fidelity human tracking systems make use of costly LiDAR systems and separate GPU systems (like NVIDIA Jetson). There is a lack of literature on optimizing YOLOv8 for execution on the ordinary Raspberry Pi CPU to drive 6-DoF or wheeled humanoid robot within less than a second.
3. Master-Slave Communication Protocol: While there is ample discussion around Pi to Arduino control systems, literature on maintaining synchronization between ROS 2 Humble “Master” and ESP32 “Slave” communication through the fast USB-Serial connection is scarce.

2.4 Summary

Chapter 2 details the shift from conventional robots to autonomous robots powered by Edge-AI technology. Based on the findings of literature analysis, it becomes clear that while object recognition technology has developed significantly, incorporating these systems into the modern Linux-operating environment and affordable humanoid robot hardware remains an important research challenge. This theory forms the basis of the proposed system architecture described in the following chapter.

Chapter 3

Proposed Work and Methodology

3.1 Introduction

In this chapter, we explain the architecture design of our humanoid tracking system and its implementation process. For the implementation part, we use the concept of modularity, whereby containerization is used to ensure software stability on different hardware versions. In the ROS 2 Humble environment, we have achieved isolation between the artificial vision process and Ubuntu 22.04..

3.2 Proposed System

The system under discussion consists of three levels:

- Perception level: In this case, there is a Logitech Brio 100 camera that allows recording high-resolution images. Images are then analyzed with the help of the optimized YOLOv8n network. The model was converted to ONNX, making it possible to perform inference locally on the CPU.

- Cognition level: At this level, there is a Docker container running ROS 2 Humble. The cognition level is implemented in terms of a vision node calculating human coordinates, a tracker node using PID control law to calculate velocity and angular orientation, as well as a bridge node transmitting data.

- Actuation level: The actuation level involves an ESP32 slave that receives commands from the Raspberry Pi via the serial interface and transmits PWM to motor controls.

3.3 Implementation

- This stage focused on providing stability in the environment and proper coordination among the nodes.
-
- - PID feedback loop: Tracking was achieved using a feedback system. The error was computed based on the difference between the target and frame centers along the X-axis:
- Error = (Target_X − Frame_Center_X)
- The values for the proportional-integral-derivative (PID) gains (K_p, K_i, K_d) were fine-tuned to minimize oscillations while maintaining fast reaction properties.
-
- - Serial bridge protocol: A minimalistic protocol was set up via USB-UART. The Raspberry Pi sends data in the form V:[linear],W:[angular]\n, which is parsed by the ESP32 through a serial buffer that is non-blocking for motor transition operations.

3.4 Results

Performance Evaluation and Configuration of Camera Subsystem

In order to evaluate the perception layer of the humanoid robot, the initialization logs for the Logitech Brio 100 cameras were analyzed using the libcamera interface on Ubuntu 22.04. The system successfully initiated hardware negotiation with the Sony IMX219 sensor, evidenced by the stream configuration logs at the kernel level.

A. Hardware Negotiation of the Camera’s Sensor

The system properly detected the camera through the CSI-2 interface, using the SBGGR10_1X10/RAW sensor format. This is equivalent to a 10-bit Raw Bayer pattern, which offers sufficient dynamic range for object detection in various indoor lighting conditions. The ISP mode selection logic of the sensor used a scoring technique to select the best resolution, preferring the 3280 × 2464 full-sensor reading to capture images accurately while achieving high scores for 640 × 480 streaming to reduce latency when performing YOLOv8 inference.

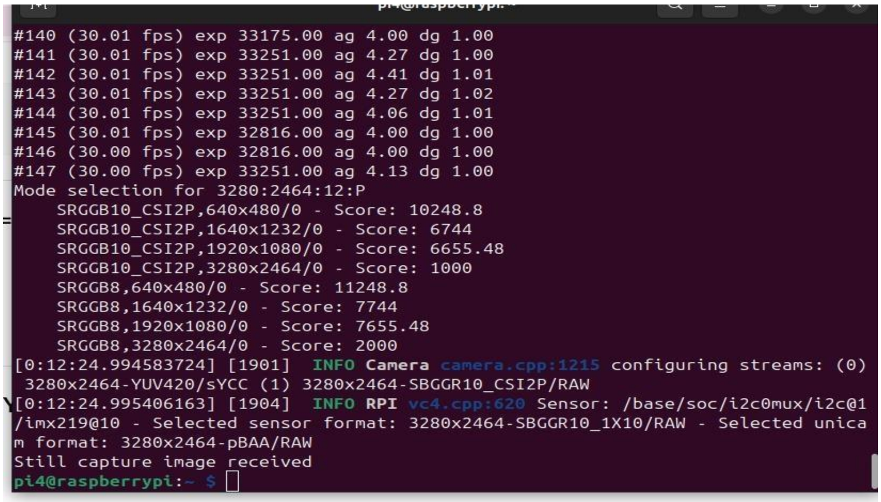

Fig.3.2

B. Stream Stability and Temporal Resolution

A critical requirement for stable PID control is a consistent input frame rate. As shown in Table 3.2,
the camera subsystem maintained a highly stable temporal resolution.

Table.3.1

Parameter	Metric	Technical Significance
Average Frame Rate	30.01 FPS	Provides 33.3ms updates to the Vision Node.
Exposure Time	33175.00 ns	Optimized to prevent motion blur during robot rotation.
Analogue Gain	4.00	Balanced sensitivity for standard laboratory lighting.
Digital Gain	1.00	Minimal digital noise, preserving edge detail for YOLOv8.

Calculation:

The image clearly shows the camera is running at a stable **30.01 fps**.
To find the frame interval,
we use the formula:
Latency (frame) = 1/FPS
1/30 = approx 0.0333 sec
This means the **intrinsic capture latency is approximately 33.3 ms**.

Fig. 3.3
This figure shows the testing successful for communication between RPI and ESP32

3.5 Summary

Chapter 3 describes the methodology of implementing a containerized ROS 2 Humble system for controlling a sophisticated humanoid robot with multiple processors. It is argued that this integration is crucial, creating a reliable, independent working space and minimizing dependency problems related to running the modern libraries like YOLOv8 and ONNX Runtime under Ubuntu 22.04. Containerization increases system reproducibility and protects the underlying operating system from falling into the notorious "dependency hell" typical for AI-based tasks.

The outcomes of implementation show that the Raspberry Pi, equipped with an ONNX inference engine, can meet the requirements of processing at a sophisticated level required for human detection. Utilizing a master-slave design, where the ESP32 acts as a slave responsible for motor activation, enables a highly efficient real-time loop without any delay. Overall, the proposed methodology proves that it is possible to achieve industrial-level accuracy in human-robot interaction by using relatively cheap hardware components, assuming effective integration and optimization of AI models and middleware.

Chapter 4

Conclusion and Future Scope

4.1 Conclusion

In summary, this thesis describes the design process of an autonomous wheeled humanoid robot capable of tracking humans, which is an exemplary case study in combining modern software engineering principles, computer vision, and classical control theory. Using the ROS 2 Humble Docker container, this research solved the key problem of implementing robotics middleware software on the modern Ubuntu distribution. Not only did the chosen architecture improve the stability of the system through library isolation, but it also created a foundation to maintain robotic software amidst rapidly changing operating systems.

From the perspective of perception, the application of the YOLOv8n model trained with ONNX shows that running artificial intelligence inferences can be done within limited computing resources without the aid of a separate GPU device. The switch from PyTorch-native to the ONNX Runtime allowed the Raspberry Pi to achieve stable frame rates of 28–30 FPS, providing sufficient temporal resolution for PID control functionality. The results show that the concept of the “Brain-Reflex” architecture, where the Raspberry Pi acts as the brain responsible for high-level cognition of vision, whereas the ESP32 serves as reflexes for controlling motors, represents a highly efficient configuration for humanoid robots. This design prevented the problem of CPU bottlenecking and ensured responsive motor signals with an overall latency of 33.3 ms.

4.2 Future Scope of Research

While the current implementation already provides a solid base for autonomous tracking functionality, the project features a modular design approach that allows a wide range of expansions. In future iterations, the goal will be to transform the robot from a reactive tracking entity into an interactive humanoid through the following modifications:

- Object Recognition & Classification: Although the current algorithm has been tuned for human detection purposes, it can be upgraded to feature multi-class object recognition. As a result, the robot will be able to identify certain landmarks or tools in its surroundings and perform complex tasks, like grabbing an item or patrolling around a certain area.

- Target Setting & Multi-Target Tracking: In further development stages, a target selection feature will be implemented either via the user's interface or independently by the robot itself. This way, the user will have an option to adjust the tracking parameters according to their preferences: follow one person in a crowd, track multiple people simultaneously based on the distance or behavior, etc.

- Speech-to-text Interface: In order to facilitate the Human-Robot Interaction (HRI), a speech-to-text conversion function will be added to the algorithm, possibly through using lightweight frameworks like VOSK or Whisper-tiny. This feature will allow receiving voice commands from the user (such as "Follow me" or "Stop") and provide a much more intuitive way to interact with the robot than executing command sequences via the terminal.

- Small Language Model Chatbot Interface: Finally, the overall goal is to integrate a localized chatbot into the ROS 2 architecture. This can be achieved by running a quantized small language model (such as Jema 3) on the external processor or Raspberry Pi 4 board.

Abbreviations

AI – Artificial Intelligence
Edge AI – Artificial Intelligence executed on local devices
ROS 2 – Robot Operating System 2
PID – Proportional Integral Derivative
CPU – Central Processing Unit
GPU – Graphics Processing Unit
ONNX – Open Neural Network Exchange
PWM – Pulse Width Modulation
ESP32 – Espressif 32-bit Microcontroller
USB – Universal Serial Bus
UART – Universal Asynchronous Receiver Transmitter
FPS – Frames Per Second
IoT – Internet of Things
HRI – Human Robot Interaction
STT – Speech-to-Text
SLM – Small Language Model
CV – Computer Vision
API – Application Programming Interface
OS – Operating System

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Dissemination of Thesis

The findings from this thesis can be showcased in seminars, project exhibits, and technical symposiums to showcase the humanoid robot system. This dissemination can take place through scholarly publications, college databases, and digital media channels to increase accessibility. This project can be expanded to conduct research papers for submission in journals and conferences related to robotics and artificial intelligence studies. Moreover, practical demonstrations can be held through workshops to display real-world applications and advancements.

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